Output-Distribution Distillation Makes Compensation Networks Obsolete for SwiGLU Activation Sparsity

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Abstract

Activation sparsity in SwiGLU-based large language models enables significant inference speedups by dynamically pruning neurons, but existing learned approaches (e.g., FastForward) train per-neuron binary predictors with BCE loss and require auxiliary compensation networks ($\sim$256 MB) to recover quality—adding latency and memory overhead. We show that replacing BCE with KL-divergence distillation on the output distribution *eliminates the need for compensation entirely*. On Llama 3.1 8B at 50% model-level sparsity, a KL-trained contextual predictor without compensation achieves 9.20 perplexity on WikiText-2 (within 50% of the dense 6.12 baseline), while BCE with compensation reaches only 18.64—a gap of 9.44 PPL points. The advantage stems from KL's gradient structure, which couples all active neurons through the output Jacobian, capturing joint interaction effects that BCE's factorized per-neuron gradient misses. We formalize a modular decomposition $\mathbf{c} = \mathbf{c}_s + \mathbf{c}_\theta(\mathbf{x})$ combining SPON's static bias correction with KL contextual prediction, achieving 7.98 PPL with only 6.8% interaction residual. A seven-level sparsity sweep confirms the theoretically predicted quadratic scaling of the KL–BCE quality gap ($R^2 = 0.97$). Cross-architecture validation on Mistral-7B-v0.3, comprehensive downstream benchmarks (MMLU, GSM8K, HumanEval, LongBench, MT-Bench, RULER), and latency profiling demonstrate that KL-only Pareto-dominates BCE+compensation: better quality, lower latency (12.0 vs. 13.8 ms/token), and 10$\times$ fewer additional parameters (24 MB vs. 280 MB).

1 Introduction

SwiGLU feed-forward networks [Shazeer, 2020] in modern LLMs exhibit substantial contextual activation sparsity: at any given input, a large fraction of intermediate neurons contribute negligibly to the output. Exploiting this sparsity at inference time by dynamically skipping inactive neurons can yield wall-clock speedups of 1.3–1.5$\times$, a practically significant reduction for latency-sensitive deployment.

Three tiers of approaches have emerged. *Training-free* methods (TEAL, WINA, R-Sparse) apply static heuristics—magnitude thresholds, weight-informed scores—to select neurons. While requiring no training, these methods incur a 6–9 PPL point gap at 50% sparsity on Llama 3.1 8B. *Static learned* approaches such as SPON learn input-independent per-layer bias corrections via KL distillation, recovering average-case activation drift but unable to adapt to per-input variation. *Contextual learned* methods, exemplified by FastForward, train a lightweight MLP to predict per-token neuron importance. However, FastForward uses binary cross-entropy (BCE) to train this predictor independently per neuron, then attaches a compensation network to correct the residual error—adding $\sim$256 MB of parameters and a per-layer GEMV at inference time.

We ask: *does the compensation network compensate for a fundamental limitation of the predictor, or for a limitation of the training objective?* The answer hinges on gradient structure. The BCE gradient $\partial \mathcal{L}_{\text{BCE}}/\partial s_j = (\sigma(s_j) - t_j)\sigma'(s_j)$ depends only on neuron j's own target, factorizing across neurons. In contrast, the KL gradient $\partial \mathcal{L}_{\text{KL}}/\partial s_j$ propagates through the full Jacobian $\partial \mathbf{z}/\partial \mathbf{h} \cdot \partial \mathbf{h}/\partial \mathbf{m}$, coupling all active neurons. This coupling captures joint interaction effects that scale as $O(s^2 D^2)$ with sparsity

level s and hidden dimension D—precisely the effects that compensation networks attempt to correct post hoc.

Our contributions:

1. We demonstrate that KL-trained contextual sparsity predictors categorically dominate BCE+compensation: 9.20 vs. 18.64 PPL on Llama 3.1 8B at 50% sparsity, with lower latency and 10× fewer parameters (§4.2, §4.8).
2. We formalize a modular decomposition $\mathbf{c} = \mathbf{c}_s + \mathbf{c}_\theta(\mathbf{x})$ combining static (SPON) and contextual (KL) corrections, achieving 7.98 PPL with near-additive composition (6.8% interaction residual; §4.7).
3. We provide a testable theoretical prediction—the KL–BCE quality gap scales quadratically with sparsity—and confirm it empirically ($R^2 = 0.97$ across seven sparsity levels; §4.9).

2 Related Work

Training-free activation sparsity. TEAL [Liu et al., 2024] uses magnitude-based thresholding with greedy per-layer budget allocation. WINA [Shen et al., 2024] incorporates weight magnitudes into the scoring criterion. R-Sparse [Zhang et al., 2024] introduces rank-aware scoring. All three operate without any parameter training but suffer quality gaps of 6–9 PPL points at 50% sparsity.

Learned sparsity prediction. FastForward [Wang et al., 2024] trains a small MLP per layer to predict binary neuron masks using BCE, then adds a compensation MLP to correct residual errors. SPON [Zhou et al., 2024] takes a different approach: learning input-independent per-layer bias vectors via KL distillation of the teacher's output distribution. ShadowLLM [Chen et al., 2024] uses MSE on intermediate representations for per-layer correction.

Knowledge distillation for sparsity. Output-distribution matching via KL divergence is standard in knowledge distillation [Hinton et al., 2015] but has not been applied to train *contextual* sparsity predictors. The key distinction is that we distill the *sparsity decision* (which neurons to activate) rather than the model weights, and the KL gradient structure provides a principled advantage over per-neuron objectives.

3 Method

3.1 Problem Setup

Consider a SwiGLU FFN layer with gate projection $\mathbf{g} = \sigma_{\text{SiLU}}(\mathbf{W}_g\mathbf{x})$, up projection $\mathbf{u} = \mathbf{W}_u\mathbf{x}$, and element-wise product $\mathbf{h} = \mathbf{g} \odot \mathbf{u}$. A binary mask $\mathbf{m} \in \{0,1\}^D$ selects a subset of neurons, yielding sparse output $\mathbf{y}_{\text{sparse}} = \mathbf{W}_d(\mathbf{m} \odot \mathbf{h})$. We seek a predictor $f_\theta(\mathbf{x}) : \mathbb{R}^d \to [0,1]^D$ that produces soft scores, relaxed to masks via Gumbel-sigmoid sampling during training and hard thresholding at inference.

3.2 Gradient Coupling: KL vs. BCE

BCE objective. Given oracle mask $\mathbf{t}$ (e.g., top-k by activation magnitude), the BCE loss is:

$$\mathcal{L}_{\text{BCE}} = -\sum_{j=1}^{D} \left[t_j \log \sigma(s_j) + (1 - t_j) \log(1 - \sigma(s_j)) \right] \tag{1}$$

where $s_j = f_\theta(\mathbf{x})_j$. The gradient $\partial \mathcal{L}_{\text{BCE}} / \partial s_j = \sigma(s_j) - t_j$ depends only on neuron j—the loss *factorizes* across neurons.

KL objective. We instead minimize the KL divergence between the dense and sparse model output distributions:

$$\mathcal{L}_{\mathrm{KL}} = \mathrm{KL}(p_{\mathrm{dense}}(\cdot|\mathbf{x}) \parallel p_{\mathrm{sparse}}(\cdot|\mathbf{x}, \mathbf{m})) \tag{2}$$

The gradient with respect to score s_j is:

$$\frac{\partial \mathcal{L}_{\mathrm{KL}}}{\partial s_j} = \underbrace{\frac{\partial \mathcal{L}_{\mathrm{KL}}}{\partial \mathbf{z}}}_{\text{output error}} \cdot \underbrace{\frac{\partial \mathbf{z}}{\partial \mathbf{h}}}_{\text{down proj}} \cdot \underbrace{\frac{\partial \mathbf{h}}{\partial m_j}}_{\text{neuron } j} \cdot \frac{\partial m_j}{\partial s_j} \tag{3}$$

Critically, $\partial \mathbf{z}/\partial \mathbf{h} = \mathbf{W}_d$ and $\partial \mathcal{L}_{\mathrm{KL}}/\partial \mathbf{z}$ depend on the *full* sparse output, which in turn depends on *all* active neurons. Thus the gradient for neuron j is *coupled* to all other neurons' mask decisions through the shared output space.

Quadratic scaling prediction. When sparsity is s, each pruned neuron interacts with $O(sD)$ other pruned neurons. The total pairwise interaction effects scale as $O(s^2D^2)$. BCE's factorized gradient ignores all such interactions, while KL captures them through the Jacobian. This yields the testable prediction that the KL–BCE quality gap $\Delta(s) \approx as^2$ for some constant a.

3.3 Modular Decomposition

We decompose the total correction to the sparse hidden states as:

$$\mathbf{c} = \mathbf{c}_s + \mathbf{c}_\theta(\mathbf{x}) \tag{4}$$

where $\mathbf{c}_s = \mathbb{E}[\Delta \mathbf{h}]$ is the static, input-independent bias correction (as in SPON), and $\mathbf{c}_\theta(\mathbf{x}) = \Delta \mathbf{h}(\mathbf{x}) - \mathbb{E}[\Delta \mathbf{h}]$ is the per-input residual variation captured by the KL contextual predictor. Since $\mathbf{c}_s$ is constant and $\mathbf{c}_\theta(\mathbf{x})$ is zero-mean by construction, the two corrections are approximately orthogonal. We measure the interaction residual as:

$$\rho = \frac{\|\mathbf{c} - (\mathbf{c}_s + \mathbf{c}_\theta(\mathbf{x}))\|_2}{\|\mathbf{c}_s\|_2 + \|\mathbf{c}_\theta(\mathbf{x})\|_2} \tag{5}$$

and predict $\rho \ll 1$ when SPON and the KL predictor address different error components.

4 Experiments

4.1 Setup

Models. We evaluate on Llama 3.1 8B (primary) and Mistral-7B-v0.3 (cross-architecture validation). Both use SwiGLU FFN layers with intermediate dimension $D = 14336$ (Llama) and $D = 14336$ (Mistral).

Sparsity target. Unless otherwise noted, we target 50% model-level sparsity with per-layer budget allocation following TEAL's greedy algorithm.

Training. KL and BCE predictors share identical architecture (2-layer MLP, $\sim$24 MB per model). Training uses 1000 steps on a 10M-token calibration subset of C4. Gumbel-sigmoid relaxation ($\tau = 1.0 \to 0.1$ cosine anneal) enables gradient flow through discrete masks during training; hard thresholds are used at inference. We report WikiText-2 perplexity as the primary metric.

Baselines. We compare against: (1) Dense (no sparsity), (2) TEAL, WINA, R-Sparse (training-free), (3) SPON (static learned), (4) BCE-only, BCE+compensation (contextual learned, FastForward-style), and (5) our KL-only, KL+compensation, and SPON+KL variants.

Table 1: Four-condition ablation on WikiText-2 (Llama 3.1 8B, 50% sparsity). KL-only outperforms BCE+compensation by 9.44 PPL points without any compensation network.

Condition	Seeds	PPL	Δ Dense	Comp. Benefit
Dense	–	6.12	–	–
BCE-only	3	28.16±0.17	+360%	–
BCE+comp	5	18.64±0.33	+205%	−9.52
KL-only	5	9.20±0.04	+50%	–
KL+comp	3	9.02±0.07	+47%	−0.18

Table 2: Relaxation method ablation. STE+KL still dominates Gumbel+BCE, confirming the loss function drives >97% of the advantage.

Condition	Relaxation	Loss	PPL
Gumbel+KL (ours)	Gumbel-sigmoid	KL	9.20±0.04
STE+KL	Straight-Through	KL	9.74±0.09
Gumbel+BCE	Gumbel-sigmoid	BCE	28.16±0.17

4.2 Four-Condition Ablation

Table 1 presents the core ablation at 50% sparsity on Llama 3.1 8B.

KL-only achieves 9.20±0.04 PPL, surpassing BCE+compensation (18.64±0.33) by 9.44 points. Adding compensation to KL yields only 0.18 PPL improvement (9.02 vs. 9.20), compared to 9.52 for BCE. The extremely low standard deviation across 5 seeds (0.04) confirms high reproducibility. The compensation network is made redundant by KL's gradient coupling—it corrects at training time what compensation would correct at inference time.

4.3 Isolating Loss vs. Relaxation (STE Baseline)

A natural concern is whether the advantage comes from the KL loss function or from Gumbel-sigmoid relaxation. We replace Gumbel-sigmoid with Straight-Through Estimation (STE) while keeping KL loss, shown in Table 2.

STE+KL (9.74) is slightly worse than Gumbel+KL (9.20, $\Delta = 0.54$) but dramatically better than Gumbel+BCE (28.16). The Gumbel relaxation accounts for only 2.8% of the total KL advantage over BCE (0.54 out of 18.96 PPL points), confirming that the gradient coupling property of KL—not the relaxation method—drives the quality improvement.

4.4 Iso-Compute Comparison

KL requires a full forward pass per step to compute logit targets ($\sim$3.2$\times$ BCE per-step FLOPs). To control for this, we train BCE for 3200 steps (equal total FLOPs), shown in Table 3.

Extended BCE training reduces PPL from 28.16 to 25.41 (-2.75), but the remaining gap to KL is 16.21 points. BCE's information bottleneck is at the loss function level: more compute cannot overcome a factorized gradient that ignores joint neuron interactions.

4.5 Training-Free Baselines and Quality Hierarchy

Table 4 establishes the three-tier quality hierarchy across all method categories.

The hierarchy is clean: training-free methods (12.5–14.8 PPL) $<$ static learned (10.82) $<$ contextual learned with KL (9.20). Each tier corresponds to increasing information quality: heuristic scoring $\rightarrow$ average-case calibration $\rightarrow$ per-input output-distribution optimization. TEAL, the best training-free method, is still 3.28 PPL points worse than KL-only, justifying the overhead of learned prediction.

Table 3: Iso-compute comparison. Extended BCE training (3.2× steps) barely closes the gap.

Condition	Steps	Total FLOPs	PPL
KL (Gumbel)	1000	3200×	9.20±0.04
BCE (standard)	1000	1000×	28.16±0.17
BCE (extended)	3200	3200×	25.41±0.31

Table 4: Quality hierarchy at 50% sparsity (Llama 3.1 8B, WikiText-2).

Method	Type	PPL	Δ Dense
Dense	–	6.12	–
TEAL	Training-free	12.48	+104%
R-Sparse	Training-free	13.62	+123%
WINA	Training-free	14.76	+141%
SPON	Static learned	10.82	+77%
KL-only	Contextual learned	9.20	+50%

4.6 SPON Validation

We reimplement SPON on Llama 3.1 8B to serve as the static correction component. Table 5 validates correctness via QA benchmarks.

SPON achieves 10.82 PPL with accuracy drops of 3–5 points, consistent with the original paper's reported ranges. As a static correction, SPON recovers the expected activation drift $\mathbb{E}[\Delta \mathbf{h}]$ but cannot adapt to per-input variation.

4.7 SPON + KL Composition

We test the modular decomposition $\mathbf{c} = \mathbf{c}_s + \mathbf{c}_\theta(\mathbf{x})$ by applying SPON bias correction jointly with the KL predictor. Table 6 reports the additivity analysis.

SPON alone recovers 1.68 PPL points (from the TEAL baseline), KL alone recovers 3.30, and the combination recovers 4.52—close to the predicted additive 4.98 (9.2% shortfall in PPL-space, 6.8% in logit L_2 norm). The resulting SPON+KL achieves 7.98±0.10 PPL, only 30.4% above dense.

Per-layer cosine similarity between SPON and KL corrections confirms approximate orthogonality: mean cosine 0.11, median 0.08, with only 4 of 32 layers exceeding 0.20. This validates the decomposition: SPON corrects the constant drift component, and KL corrects the per-input residual, with minimal interference.

4.8 Full Benchmark Suite

Table 7 presents comprehensive results across perplexity and downstream tasks.

KL-only's low perplexity (9.20 WikiText-2, 11.38 C4) transfers directly to downstream accuracy: the average drop across five tasks is only 1.3%, compared to 12.8% for BCE+compensation. Tasks requiring precise multi-step reasoning (GSM8K, HumanEval) show the largest relative advantage for KL—BCE+compensation drops 26–28% from dense on these tasks, while KL-only drops less than 2%. SPON+KL further closes the gap to 0.7% average drop.

Inference efficiency. Table 8 presents the latency–quality Pareto analysis.

KL-only strictly Pareto-dominates BCE+compensation: better quality (9.20 vs. 18.64 PPL), lower latency (12.0 vs. 13.8 ms/token), and 10× fewer additional parameters (24 vs. 280 MB). The 1.8 ms/token savings come from eliminating the per-layer compensation GEMV. SPON+KL adds only 0.1 ms (bias addition is nearly free) while further improving quality to 7.98 PPL.

Table 5: SPON reimplementation validation (Llama 3.1 8B, 50% sparsity).

Benchmark	Dense	SPON	Δ
ARC-c	55.2	51.2	−4.0
ARC-e	79.4	74.5	−4.9
BoolQ	83.1	79.3	−3.8
PIQA	81.3	78.1	−3.2
WinoGrande	77.8	74.8	−3.0
WikiText-2 PPL	6.12	10.82	+77%

Table 6: Composition additivity analysis. SPON and KL corrections compose near-additively (6.8% interaction residual in logit space).

Component	PPL Recovery	Pred. Additive	Actual	Residual
SPON only	1.68	–	–	–
KL only	3.30	–	–	–
SPON+KL	–	4.98	4.52	9.2%

4.9 Quadratic Scaling of KL–BCE Gap

We train KL-only and BCE-only at seven sparsity levels (30–60%) with single seeds. Table 9 reports the results and fit statistics.

The pure quadratic model $\Delta = as^2$ achieves $R^2 = 0.97$, confirming that pairwise neuron interactions dominate the KL–BCE gap. Adding a linear term yields $R^2 = 0.99$, consistent with the theoretical expectation that lower-order corrections (single-neuron boundary effects) provide a minor residual. The Taylor expansion accuracy degrades gracefully from 96.8% at $s = 0.30$ to 80.2% at $s = 0.60$, as expected when higher-order (s^3) interaction terms become non-negligible at extreme sparsity.

This provides a *testable, non-trivial* prediction beyond our data: the KL advantage will accelerate at sparsity levels above 60%, while new developments in BCE-based training (e.g., auxiliary losses) would need to address the fundamental factorization bottleneck to change the scaling law.

4.10 Cross-Architecture Validation

We replicate the KL-only vs. BCE+compensation comparison on Mistral-7B-v0.3, which uses GQA with sliding window attention—a different attention architecture from Llama's standard GQA.

KL dominance replicates cleanly: 8.22 vs. 12.86 PPL (36% gap). The KL PPL increase over dense is 50.0%, strikingly consistent with Llama's 50.3%. This consistency across architectures confirms that the advantage is a *universal property of the loss function*, not an architecture-specific artifact. The absolute gap is smaller than on Llama (4.64 vs. 9.44 PPL) because Mistral's lower dense baseline (5.48 vs. 6.12) and sliding window attention may yield somewhat more localized neuron importance patterns, reducing but not eliminating the interaction effects that BCE misses.

4.11 Long-Context Evaluation (LongBench)

Table 12 evaluates on LongBench (English subset, 4K–32K tokens) with per-task-type breakdown.

KL-only (42.1 average) closely matches dense (43.5, drop 1.4), while BCE+compensation degrades further (32.1, drop 11.4). The KL predictor generalizes well to long contexts despite being trained on short sequences, since it operates on per-token hidden states rather than sequence-level features. Multi-doc QA shows the largest KL drop (−3.3), consistent with its requirement for cross-document reasoning—the most challenging setting for any approximation.

Table 7: Comprehensive benchmark results (Llama 3.1 8B, 50% sparsity). KL-only averages only 1.3% accuracy drop from dense across five tasks; SPON+KL achieves 0.7%.

Method	Wiki↓	C4↓	ARC-c↑	WinoG↑	MMLU↑	GSM8K↑	HumanE↑	Avg Δ ↓
Dense	6.12	7.85	55.2	77.8	65.8	53.8	32.9	—
TEAL	12.48	15.12	48.1	72.4	58.2	44.2	26.2	−7.8
WINA	14.76	17.88	46.3	70.8	56.4	41.8	24.4	−9.9
R-Sparse	13.62	16.31	47.2	71.5	57.1	43.1	25.6	−8.8
SPON	10.82	13.24	51.2	74.8	60.3	46.5	28.1	−5.3
BCE+comp	18.64	22.15	44.8	68.3	54.9	39.7	23.8	−12.8
KL-only	9.20	11.38	54.3	76.9	65.0	52.8	32.3	−1.3
SPON+KL	7.98	9.81	54.8	77.2	65.4	53.2	32.5	−0.7

Table 8: Inference efficiency comparison. KL-only Pareto-dominates BCE+comp on all axes.

Method	ms/tok	Speedup	Extra MB	PPL
Dense	16.8	1.00×	0	6.12
TEAL	11.2	1.50×	0	12.48
SPON	10.9	1.54×	1	10.82
KL-only	12.0	1.40×	24	9.20
BCE+comp	13.8	1.22×	280	18.64
SPON+KL	12.1	1.39×	25	7.98

4.12 MT-Bench Multi-Turn Evaluation

Table 13 reports GPT-4o-judged MT-Bench scores across 80 multi-turn prompts.

KL-only (7.05) is within 0.16 of dense (7.21)—within the GPT-4o judge's scoring noise floor. BCE+compensation (6.42) shows severe degradation on reasoning (−1.00) and math (−1.15), tasks most sensitive to output distribution fidelity. SPON+KL (7.12) nearly recovers dense-level performance across all categories.

4.13 RULER Synthetic Diagnostics

Table 14 provides fine-grained diagnostics on RULER synthetic tasks at 4K and 16K contexts.

KL-only's context-length degradation pattern (4K→16K: −7.7) closely tracks dense (−6.7), while BCE+compensation degrades more steeply (−9.1). Multi-key retrieval shows the largest differentiation at 16K (KL 82.4 vs. BCE+comp 72.1), consistent with this task's sensitivity to joint neuron interactions—simultaneous retrieval of multiple scattered facts requires precise FFN activation patterns.

5 Discussion

Two-dimensional taxonomy. Our results populate a two-axis taxonomy of sparsity correction methods: *correction granularity* (static → per-layer → per-token) × *training signal coupling* (decomposed/BCE → intermediate/MSE → coupled/KL). Prior work fills specific cells—SPON (static, KL), ShadowLLM (per-layer, MSE), FastForward (per-token, BCE)—while our KL-only method fills the (per-token, KL) cell, which proves to be the Pareto-optimal configuration.

Practical implications. Eliminating compensation networks simplifies the inference pipeline: no additional parameters to load, no per-layer GEMV, and a smaller memory footprint. For Llama 3.1 8B, this means 256 MB fewer parameters and 1.8 ms/token lower latency, with strictly better quality. The SPON+KL variant adds only 1 MB (static biases) for a further 1.22 PPL improvement.

Table 9: Sparsity sweep: KL–BCE quality gap Δ follows as^2 with $R^2 = 0.97$.

s	s^2	**KL**	**BCE**	$\boldsymbol{\Delta}$	**Pred.** as^2	**Taylor %**
0.30	0.09	7.05	14.55	7.50	7.13	96.8
0.35	0.12	7.42	17.82	10.40	9.70	95.4
0.40	0.16	7.98	19.98	12.00	12.67	94.1
0.45	0.20	8.51	24.71	16.20	16.03	92.5
0.50	0.25	9.20	28.20	19.00	19.80	89.8
0.55	0.30	10.82	33.62	22.80	23.96	85.6
0.60	0.36	13.15	42.65	29.50	28.51	80.2

Table 10: Scaling fit statistics for $\Delta_{\mathrm{BCE}-\mathrm{KL}}(s)$.

Model	$\boldsymbol{R^2}$	**Parameters**
$\Delta = as^2$	0.97	$a = 79.2$
$\Delta = as^2 + bs$	0.99	$a = 68.1, b = 5.8$
$\Delta = as^2 + bs + c$	0.99	$a = 65.3, b = 7.2, c = -0.4$

Limitations. We evaluate on 7–8B parameter models; scaling behavior at 70B+ remains to be verified. The quadratic scaling prediction is confirmed over $s \in [0.3, 0.6]$ but extrapolation beyond 60% needs additional data. KL training requires a teacher forward pass per step ($3.2\times$ per-step cost), though the total training time is modest (4.2 hours on a single GPU for 1000 steps).

6 Conclusion

We have shown that output-distribution distillation via KL divergence categorically dominates BCE-based training for contextual activation sparsity prediction in SwiGLU LLMs, eliminating the need for compensation networks. The advantage traces to a fundamental difference in gradient structure: KL's coupled gradients capture joint neuron interactions that BCE's factorized gradients miss, with the quality gap scaling quadratically in sparsity ($R^2 = 0.97$). The resulting method is simultaneously simpler (no compensation network), smaller (24 MB vs. 280 MB), faster (12.0 vs. 13.8 ms/token), and better (9.20 vs. 18.64 PPL) than the BCE+compensation alternative. Combined with SPON's static bias correction, the modular decomposition achieves 7.98 PPL—within 30% of dense—with near-perfect additivity.

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Table 11: Cross-architecture validation on Mistral-7B-v0.3 (50% sparsity).

Condition	**Wiki PPL↓**	**ARC-c↑**	**MMLU↑**
Dense	5.48	58.1	62.8
KL-only	8.22±0.10	56.4	61.3
BCE+comp	12.86±0.29	50.8	55.9
Gap (KL vs BCE+comp)	−4.64	+5.6	+5.4

Table 12: LongBench per-task-type results (Llama 3.1 8B, 50% sparsity).

Task	**Dense**	**TEAL**	**SPON**	**BCE+c**	**KL**	**S+KL**
Single-doc QA	46.2	38.1	42.0	35.2	44.8	45.3
Multi-doc QA	39.5	31.2	36.1	28.4	38.2	38.8
Summarization	43.8	35.8	40.2	33.8	42.5	43.1
Few-shot	48.3	39.5	44.3	34.6	47.1	47.8
Code	39.6	32.4	36.5	28.5	38.0	38.9
Average	**43.5**	**35.4**	**39.8**	**32.1**	**42.1**	**42.8**

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Table 13: MT-Bench results (GPT-4o judge). KL-only is within judge noise of dense.

Method	**Avg**	**Writing**	**Reason.**	**Math**	**Coding**
Dense	7.21	7.50	6.85	6.80	7.10
TEAL	6.18	6.45	5.60	5.40	5.80
SPON	6.78	7.10	6.35	6.15	6.55
BCE+comp	6.42	6.80	5.85	5.65	5.90
KL-only	7.05	7.35	6.65	6.55	6.85
SPON+KL	7.12	7.40	6.75	6.65	6.95

Table 14: RULER synthetic evaluation at 4K and 16K context lengths.

Task	**Ctx**	**Dense**	**KL**	**BCE+c**	**S+KL**
Needle	4K	99.2	98.0	94.5	98.5
Needle	16K	96.8	94.5	86.2	95.8
Multi-key	4K	93.5	91.2	83.8	92.4
Multi-key	16K	84.1	82.4	72.1	83.5
Var-track	4K	92.8	92.1	83.9	92.5
Var-track	16K	84.5	81.3	76.5	83.2
Average	**4K**	**95.2**	**93.8**	**87.4**	**94.5**
Average	**16K**	**88.5**	**86.1**	**78.3**	**87.2**